

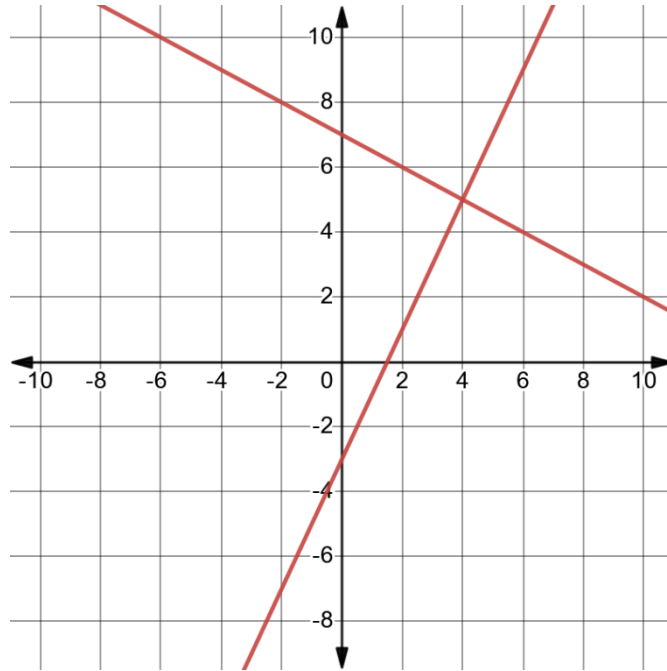
Geometry
Unit 13
Lesson 1 Practice

Name ___ **Answer Key** ___
Date _____
Period _____

1. Graph the equations on the coordinate plane:

$$y = -\frac{1}{2}x + 7$$

$$y = 2x - 3$$



2. The equations $y = -\frac{1}{2}x + 7$ and $y = 2x - 3$ resulted in _____ lines.

☐ parallel
☒ **perpendicular**
☐ vertical

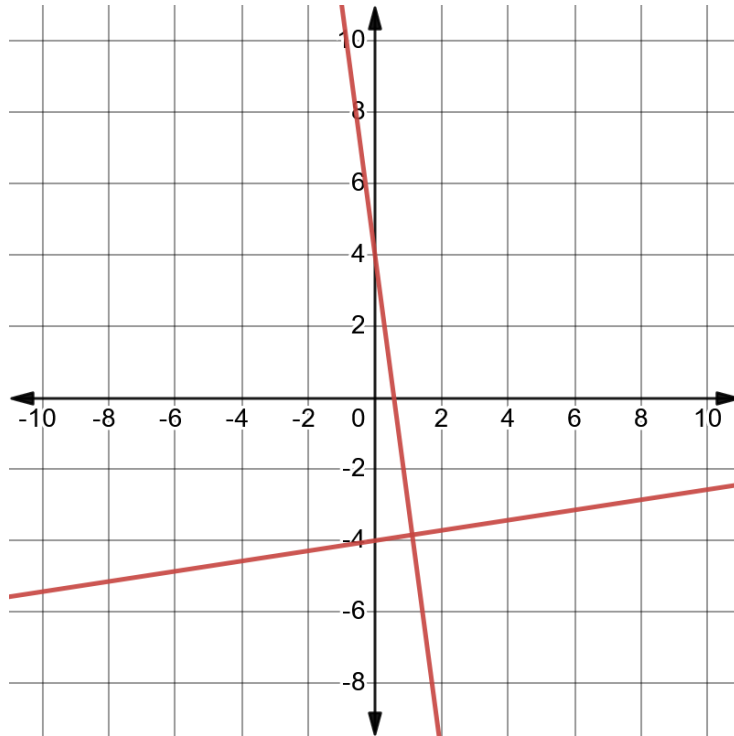
3. The slopes of the equations are _____

☐ equal.
☐ undefined.
☒ **opposite reciprocals.**

4. Rewrite the equation $7y - x = -28$ in slope intercept form and graph the equation using slope-intercept form.

$$7y = x - 28$$

$$y = \frac{1}{7}x - 4$$



5. Determine if $y = -7x + 4$ is parallel, perpendicular, or neither to the graph of the equation $7y - x = -28$.
- $$y = -7x + 4 \rightarrow m = -7$$
- $$7y - x = -28 \rightarrow m = \frac{1}{7}, \text{ so perpendicular}$$

6. Graph the equation $y = -7x + 4$ on the graph to verify your answer to question 5.
See graph.

7. Which equation represents a line which is perpendicular to the line $y = \frac{3}{8}x - 1$?

$$m = -\frac{8}{3}$$

☐ $3x + y = -40$

☐ $3x - 8y = -56$

☐ $3y - 8x = 3$

☒ $8x + 3y = 9$

$$-8y = -56 - 3x \rightarrow m = \frac{3}{8}$$

$$3y = 3 + 8x \rightarrow m = \frac{8}{3}$$

$$3y = 9 - 8x \rightarrow y = 3 - \frac{8}{3}x \rightarrow m = -\frac{8}{3}$$

8. Which equation represents a line which is parallel to the line $y = -\frac{2}{3}x + 1$?

$$m = -\frac{2}{3}$$

☐ $2x - 3y = 21$

☐ $2y - 3x = 2$

☐ $3x - 2y = -2$

☒ $2x + 3y = -21$

$$-3y = 21 - 2x \rightarrow y = -7 + \frac{2}{3}x \rightarrow m = \frac{2}{3}$$

$$2y = 2 + 3x \rightarrow m = \frac{3}{2}$$

$$-2y = -2 - 3x \rightarrow m = \frac{3}{2}$$

$$3y = -21 - 2x \rightarrow y = -7 - \frac{2}{3}x \rightarrow m = -\frac{2}{3}$$